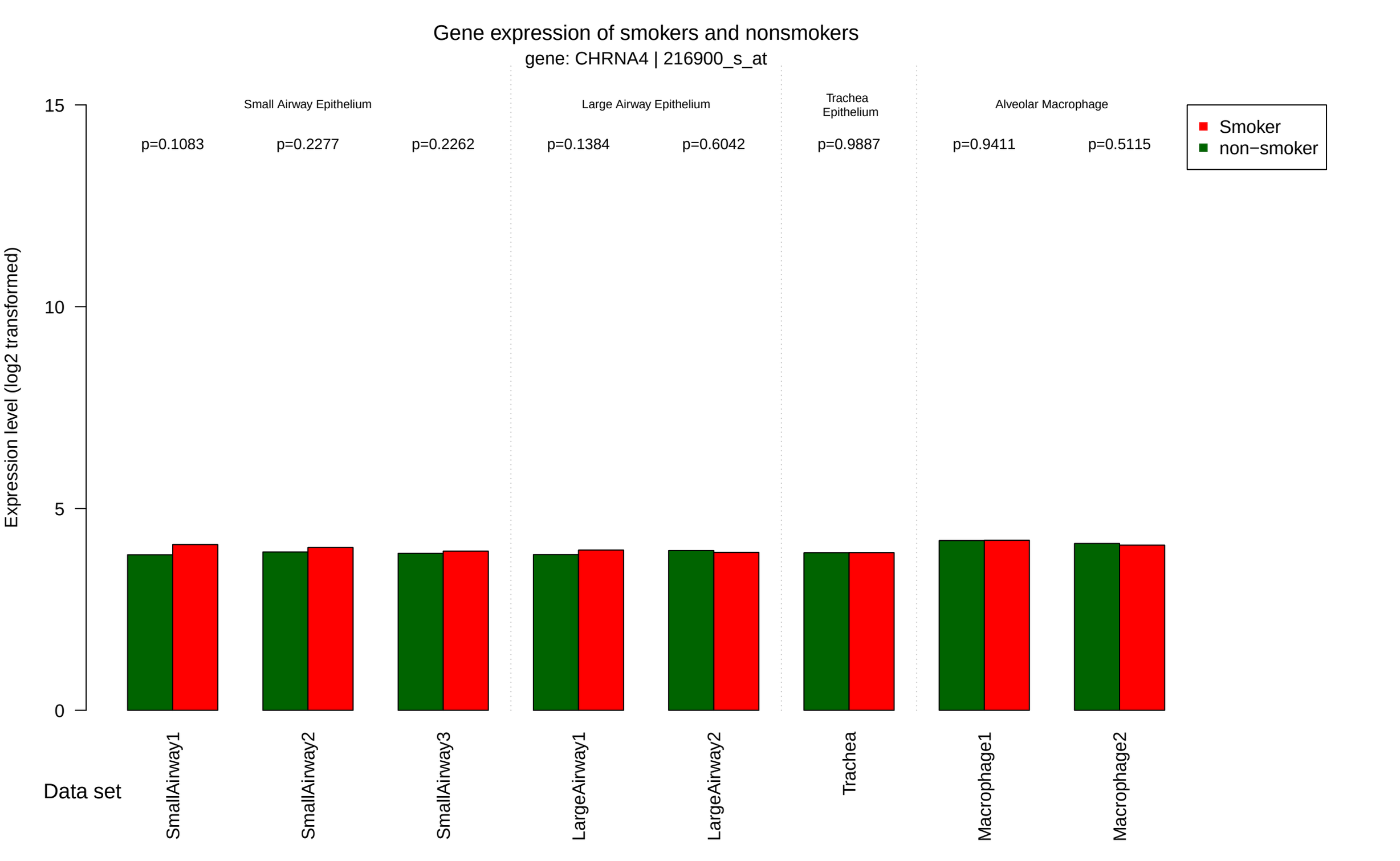
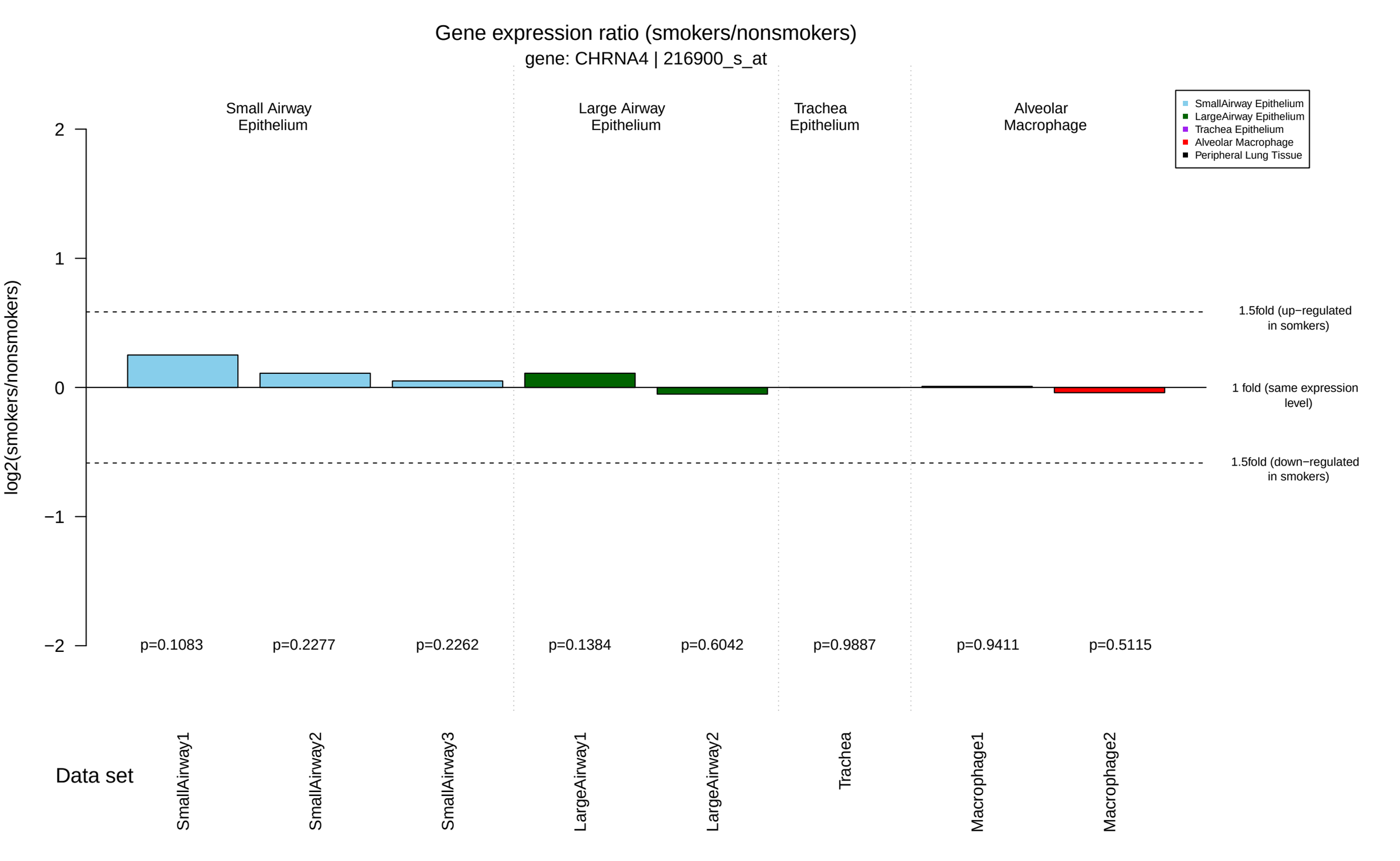
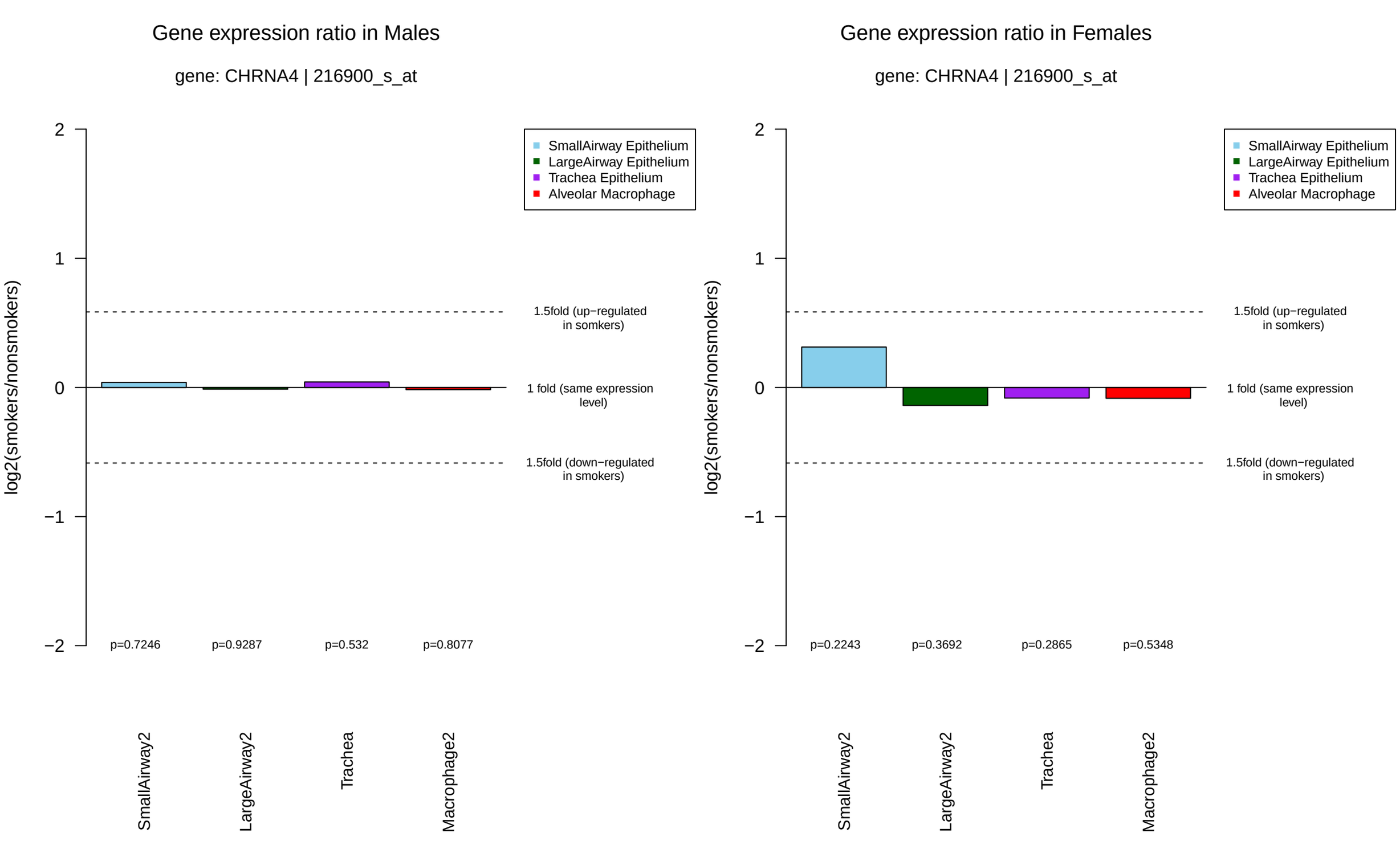
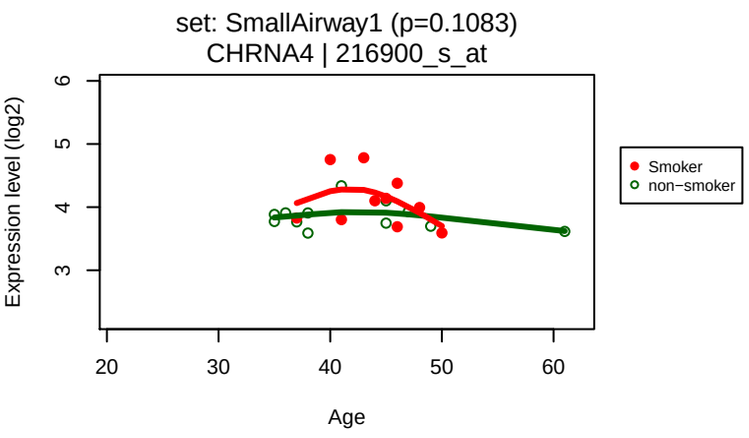
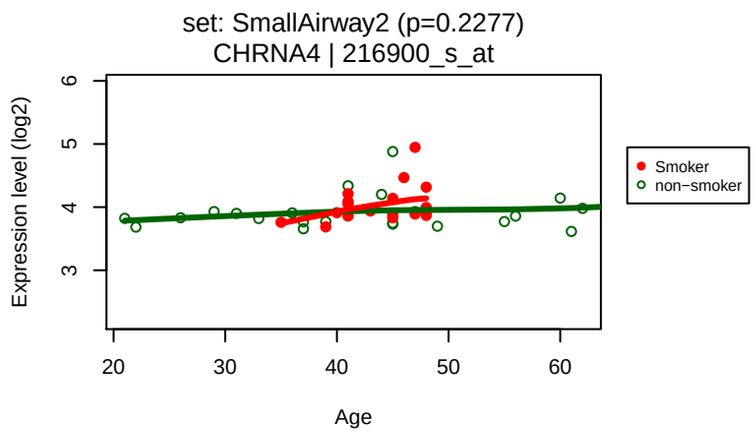
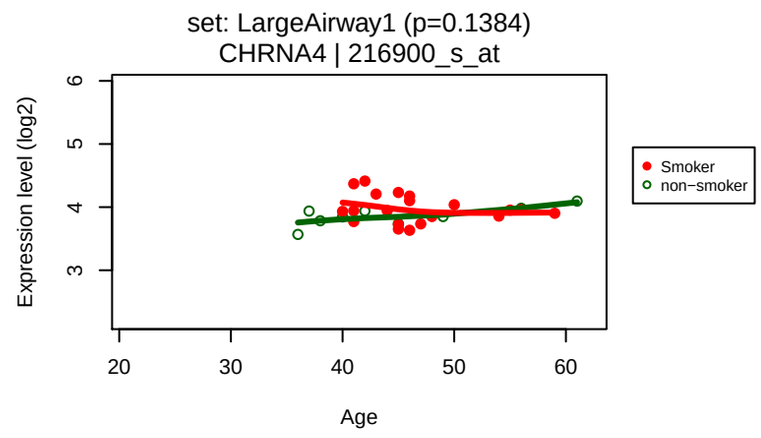

Gene expression of smokers and nonsmokers
gene: CHRNA4 | 216900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1083
p=0.2277
p=0.2262
p=0.1384
p=0.6042
p=0.9887
p=0.9411
p=0.5115
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

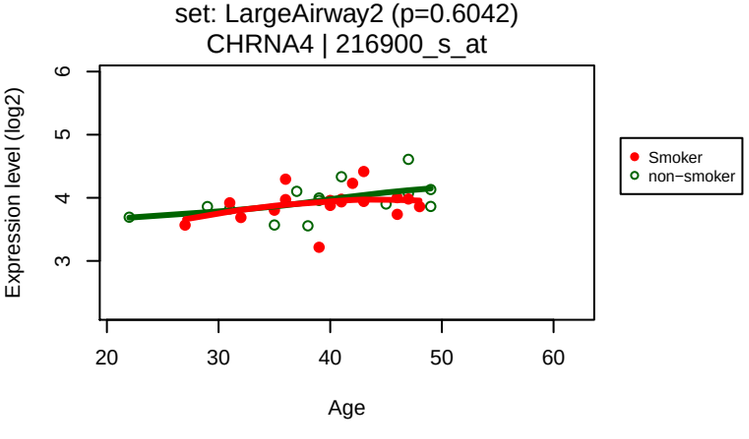

Gene expression ratio (smokers/nonsmokers)
gene: CHRNA4 | 216900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1083
p=0.2277
p=0.2262
p=0.1384
p=0.6042
p=0.9887
p=0.9411
p=0.5115
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

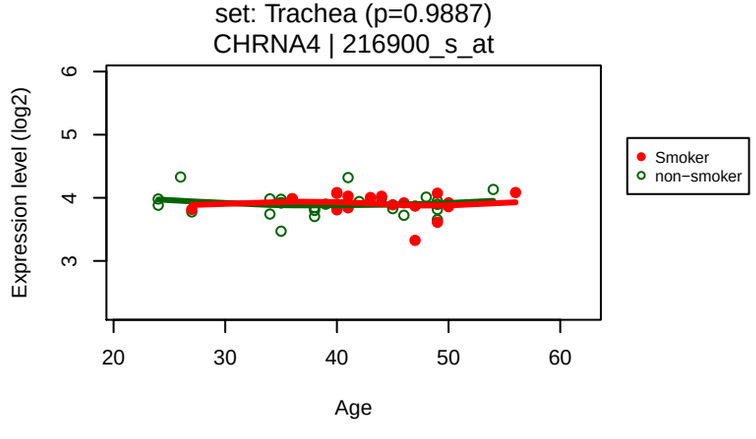

Gene expression ratio in Males
gene: CHRNA4 | 216900_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7246
p=0.9287
p=0.532
p=0.8077
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHRNA4 | 216900_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2243
p=0.3692
p=0.2865
p=0.5348
SmallAirway2
LargeAirway2
Trachea
Macrophage2

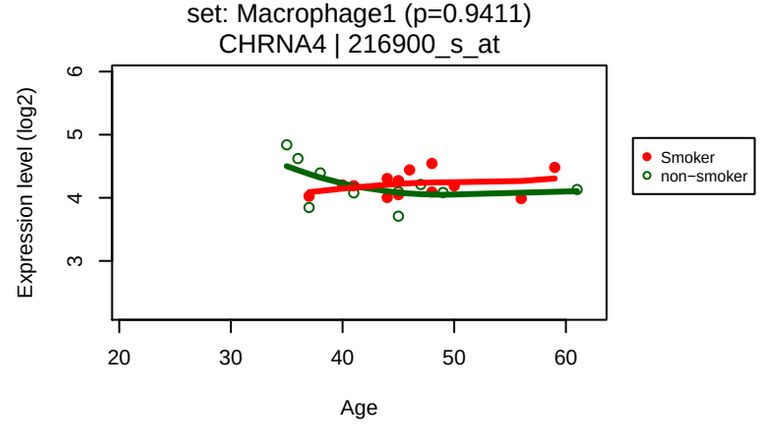

set: SmallAirway1 (p=0.1083)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker

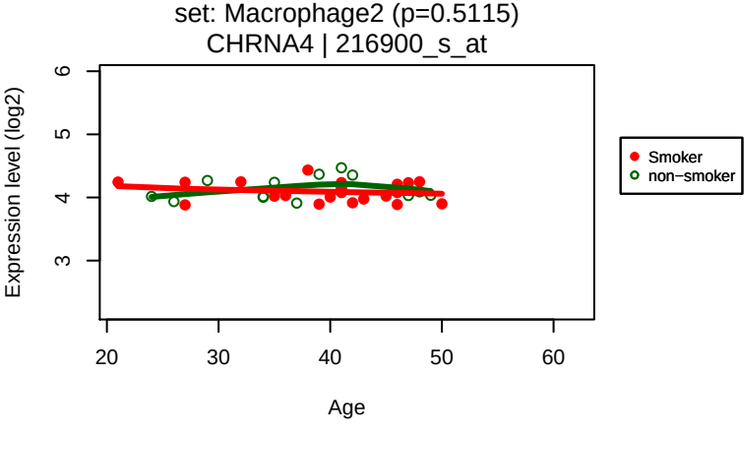

set: SmallAirway2 (p=0.2277)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker

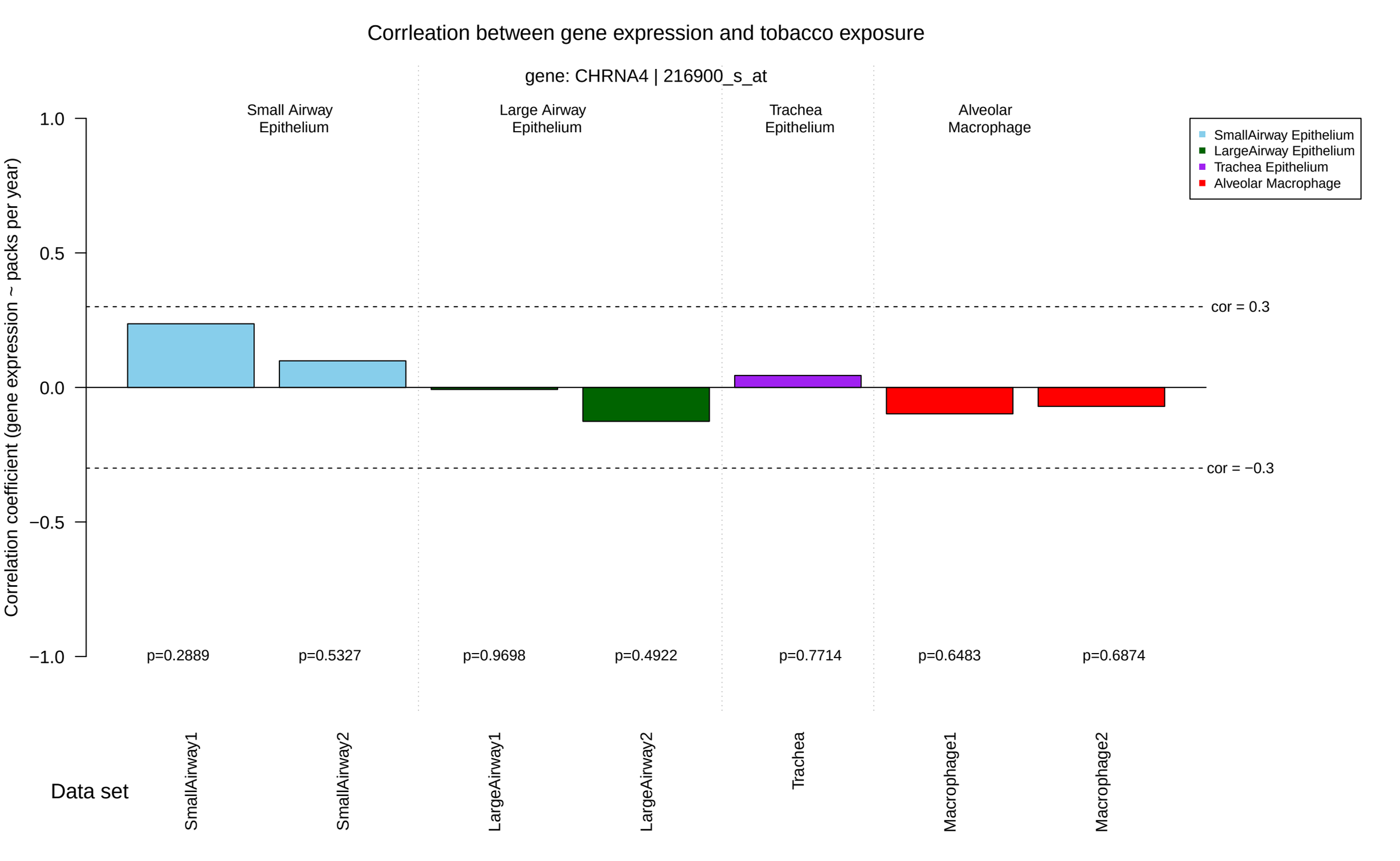

set: LargeAirway1 (p=0.1384)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6042)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9887)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9411)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.5115)
CHRNA4 | 216900_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHRNA4 | 216900_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2889
p=0.5327
p=0.9698
p=0.4922
p=0.7714
p=0.6483
p=0.6874
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2